Use Case Template

CardioAI - Heart Disease Prediction System

1. Use Case Title	Train and Cross-Validate Models
2. Abbreviated Title	Train Models
3. Use Case Id	UC-07
4. Actors	Clinician (Primary), ML Engine (Supporting)
5. Description	The Clinician initiates a model training pipeline with k-fold cross-validation on the heart disease dataset to produce a deployable CNN-LSTM-Attention model with validated performance metrics.
5.1. Pre Conditions	1. Preprocessed dataset is available. 2. Training hyperparameters are configured. 3. GPU/CPU resources are available.
5.2. Task Sequence	1. Clinician configures training parameters (epochs, learning rate, k-folds). 2. Clinician initiates training via CLI or dashboard. 3. ML Engine loads and preprocesses the dataset. 4. ML Engine performs k-fold cross-validation. 5. ML Engine trains the CNN-LSTM-Attention model on each fold. 6. System logs per-fold and average metrics (Accuracy, F1, ROC-AUC). 7. System saves the best model checkpoint.
5.3. Post Conditions	1. A trained model checkpoint is saved to disk. 2. Cross-validation metrics are logged and available for review.
6. Modification History	Date: September 11, 2026
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